

Objective: To learn the basics of estimating the time and cost for a software project, even with no prior technical knowledge.

- You will break down a client request into small tasks,
- estimate the effort for each,
- calculate a rough project cost.

Your Role: You are part of a team that has just been approached by a potential client. Your job is to create an initial rough estimate to help the client understand the potential scope and cost of their idea.

The Client Request: "Quick Cup" Coffee Ordering App

- **Client:** The owner of "The City Coffee Shop," a popular coffee shop near campus.
- **The Problem:** The shop gets extremely busy between classes, causing long lines and lost customers.
- **The Idea:** "I want a simple mobile app so students can order and pay for their drinks before they arrive. They just show an order number, grab their coffee, and go. It needs to be super simple for both the students and my baristas."

Core Feature Requirements:

1. Student View:

- See a menu of drinks and prices.
- Add drinks to a cart.
- Customize a drink (e.g., add syrup, choose milk type).
- Place an order (we'll assume payment is handled simply, like "Pay at Counter" to start).
- Receive a confirmation screen with an order number.

2. Barista View (for an in-store tablet):

- See a list of new incoming orders.
- Mark an order as "In Progress."
- Mark an order as "Ready for Pickup."

Part 1: Break It Down

The first rule of estimation is: **You can't estimate what you don't understand.** We need to break this big project into smaller, manageable pieces.

Instructions: Fill out the table below. First, identify the major components (we've given you two "Epics" to start, For now think of an Epic as a body of work that can be broken down into specific tasks). Then, for each component, list the specific features or tasks required to make it work.

HINT: Think step-by-step. What is the very first thing a user needs to see? What comes next? What about the barista? Ask yourself "What else is needed for this to work?" For example, a user needs to see the menu before they can *add an item* to the cart. These are two separate tasks!

Epic (Major Component)	Features / Tasks (The smaller pieces)
Student-Facing Mobile App	1. <i>Example: Design the Menu Screen</i>
	2.
	3.
	4.
	5.
	6. <i>...add more rows if needed</i>
Barista-Facing Tablet App	1. <i>Example: Display list of new orders</i>
	2.
	3.
	4. <i>...add more rows if needed</i>

Part 2: Estimate the Effort

In this step, you will estimate the effort required to complete each task identified in Part 1. Because software development involves uncertainty, we will use the **three-point estimation method** to represent a range of possible outcomes for each task:

- **Optimistic (O):** The minimum effort required, assuming ideal conditions and no difficulties.
- **Most Likely (M):** The realistic effort required under normal circumstances.
- **Pessimistic (P):** The maximum effort required, assuming significant obstacles or delays.

To obtain a balanced estimate, apply the **Weighted Average Formula**:

$$\text{Weighted Estimate (Hours)} = (O + 4M + P) / 6$$

This approach gives greater weight to the *Most Likely* scenario while still accounting for best- and worst-case situations.

Instructions:

1. For each task listed in Part 1, provide values for O, M, and P (in hours).
2. Use the formula above to calculate the weighted estimate.
3. Record all results in the table provided.

Example:

- A simple task such as designing a confirmation screen may be: O = 1, M = 2, P = 4 → Weighted Estimate = 2.2 hours.
- A more complex task such as implementing drink customization logic may be: O = 4, M = 8, P = 20 → Weighted Estimate = 9.3 hours.

Task Description (from Part 1)	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Weighted Estimate (Hours)
<i>Example: Design the Menu Screen</i>	2	4	8	4.3
SUBTOTAL (All Tasks)				[A] TOTAL HOURS =

Part 3: Calculate the Cost (From Time to Money)

An estimate in hours is useful, but clients need to understand the cost. Let's convert your time estimate into a project budget.

Instructions: Use your Total Hours from Part 2 to complete the calculations below.

HINT: No project goes exactly as planned. You need a contingency buffer for things you didn't think of: bug fixes, client meetings, testing, etc. A standard buffer is 20-30%. This isn't padding; it's responsible risk management.

Calculation Step	Your Numbers	Result
1. Total Estimated Hours (from Part 2)	Value of [A]	

2. Add Contingency Buffer (25%)	Multiply your Total Hours by 1.25	[B] Final Project Hours =
3. Calculate Labor Cost (Assume a rate of \$50/hour)	Multiply [B] by \$50	[C] Estimated Project Cost =

Part 4: Reflection

The final number is important, but the process of getting there is even more valuable.

Instructions: Answer the following questions based on your experience with this exercise.

1. What was the hardest part of this estimation process and why?
2. What are at least TWO major assumptions you made? (e.g., about technology, design, or features)
3. If you could ask the client three questions to make your estimate more accurate, what would they be?